FINAL EXAM

Question-1

Data preprocessing is essential as raw data is often looks messy and not suitable for modeling, because it usually have missing values, nulls, various formats so with all these unnecessary disturbances in dataset, training and modeling with the machine-learning algorithm cannot be done properly, if done we may get wrong predictions. In order to avoid these wrong predictions the dataset has to undergo preprocessing steps - it helps in cleaning, organizing and helps in proper formatting the dataset so that the model can learn and train better. so this helps training correctly with accurate outputs.

Below are the most common preprocessing techniques:

1. **Handling Missing Values:**

   Almost in every dataset we find the most common thing was missing values and null values. For modeling in machine learning - with missing values it cannot train properly because it might miss some important information from that missing dataset so to overcome this the data has to be cleaned in preprocessing steps. Usually for this cleaning we either remove the rows or it can be filled with median or mode imputation. This makes the dataset more accurate and reliable before training a model.

2. **Feature scaling:**

   Different features in the dataset may have extremely different ranges. For instance, age could range from 20 to 50, and income could be in the thousands. A few models might get confused by this difference because large-range values dominate over smaller values. In order to avoid this confusion and to make sure that each feature contributes equally, feature scaling helps in adjusting all numerical values to the same range. The model trains more effectively when techniques like standardization (mean 0, standard deviation 1) or normalization (0–1 range) are applied.

3. **Encoding Categorical Variables**

   In the dataset, some columns contain text values, such as gender - Male/Female, or any other city names. As machine-learning models do not understand text, we convert these categories to numbers. Label encoding and one-hot encoding are types of encoding technologies that convert text labels into numerical representation without affecting their meaning. This phase guarantees that the model can be correctly trained using both text and numerical data.

4. **Removing Outliers:**

   Outliers are values that are significantly higher or lower than the remaining part of the dataset. For example, if one person in a dataset makes $5 million while the majority make between $30,000 and $90,000, that number is an outlier. Outliers might confuse the model by deviating from usual averages and patterns. During preprocessing, we discover these outlier values and either eliminate or replace them with more reasonable ones. As a result, the model learns more reliable and precise patterns.

Question-2:

1. **Confusion Matrix:**
   A confusion matrix helps in evaluating the performance of a classification model by revealing how many accurate predictions the model made as well as various types of errors. It divides the findings into true positives, true negatives, false positives, and false negatives instead of giving a single score. Four useful metrics are derived from this table:

   Accuracy: Measures how accurate the model is overall. The simple question, "Out of all predictions, how many were right?" is tackled. When both classes are fairly balanced, accuracy is helpful.

   Precision: handles only the cases that the model predicted as positive. It addresses the question: "When the model predicted somebody would donate (or will default), frequently was it actually true?" When false alarms are expensive, this is essential.

   Recall: It assesses how well the model captures true positive cases. "How many actual donors (or defaulters) did the model find?" . This is important when missing a positive case is more severe than a false alarm.

   F1-Score: It Offers a balanced perspective by integrating recall and precision into one number. It is useful when you want a single metric that accounts for both types of errors, particularly in cases where the dataset is imbalanced.

   Overall, the confusion matrix is most appropriate for gaining an overall understanding of classification performance instead of relying on a single metric. It clearly shows what the model gets accurately and what are the errors it makes.

2. **RMSE (**Root mean squared error)

   It is commonly used for developing regression models and the goal is to predict a number, like donation amount, sales, price, or risk score.

It indicates the average variation among our estimated values and actual values, yet this gives more weight to larger errors.
This approach works best when:
You care about the number of your prediction errors.

Bigger errors (for example, expecting a donation of $100 when the donor actually donates just $5) are more expensive.

Because RMSE is presented in the same units as the target value (dollars, points, etc.), you require an easily understood metric.

RMSE simply illustrates how close or far your model's expectations are to reality in the real world.

3. **K-Fold Cross Validation:**

K-fold cross-validation is used to assess a model's stability and predictability.
Usually, we split the dataset into smaller chunks (folds) then train and test the model several times rather than just once. This shows us if the model works well on a single split or consistently.

<u>This approach works best when:</u>

You don't want to rely too heavily on a single train-test split.

You want to make good use of your tiny dataset.

You must fairly compare models and select the most reliable one.

K-fold validation offers a more accurate assessment of the model's performance on new, unseen data by continually evaluating the model on different parts of the data.